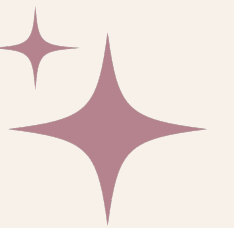
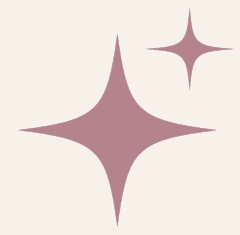


Project Idea

- **This project is designed to detect dangerous levels of Carbon Monoxide (CO) gas in the surrounding air.**
- **If the gas concentration exceeds a safe threshold, the system activates an alarm to ensure immediate safety.**
- **The system utilizes a Buzzer and LED as a dual-alert mechanism to warn users effectively.**



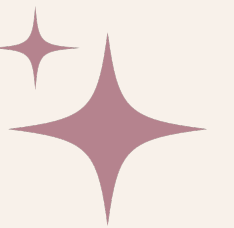


The Problem: An Invisible Threat

⚠ Toxic Nature: CO is a poisonous gas produced by incomplete combustion in heaters and stoves.

💡 The Silent Killer: It is completely colorless and odorless, making it impossible for humans to detect.

🚒 Fatal Accidents: Most injuries occur because victims cannot sense the danger before it's too late.



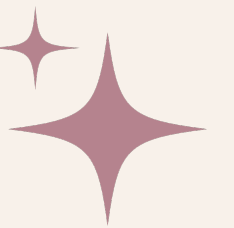


Project Objectives

🛡️ **Early Detection:** Building a high-sensitivity system to sense CO gas at its earliest stages.

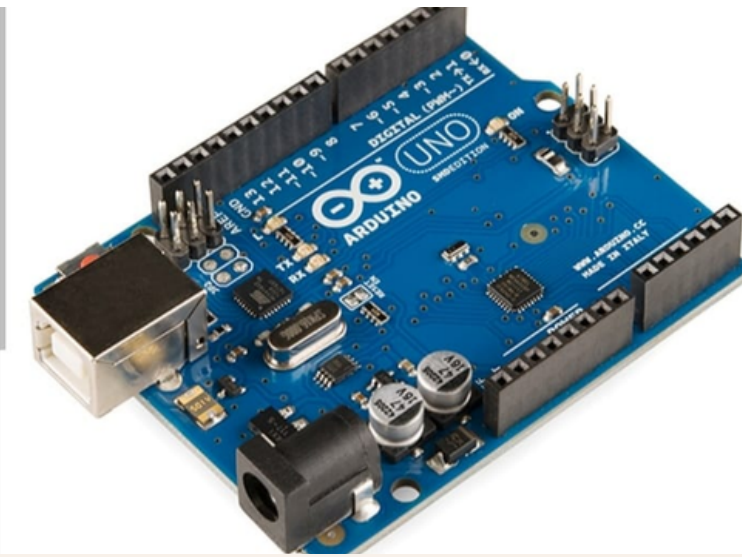
💰 **Cost-Effectiveness:** Developing an affordable, low-cost safety solution for every home.

📢 **Instant Alerting:** Ensuring user safety through immediate audio and visual warning signals.



System Components

Arduino Uno: The "Brain" of the project that processes sensor signals and executes the safety logic.

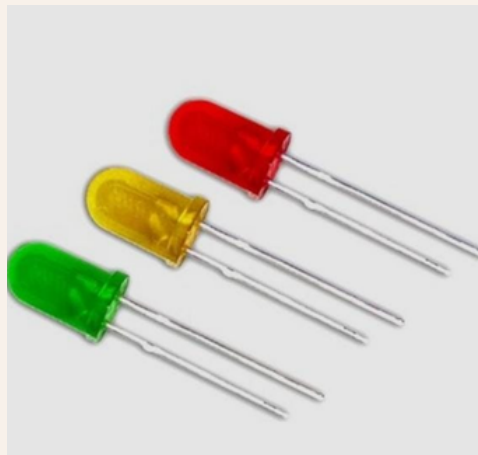


MQ-7 Sensor: The "Nose" of the project, specialized in sensing CO concentrations.



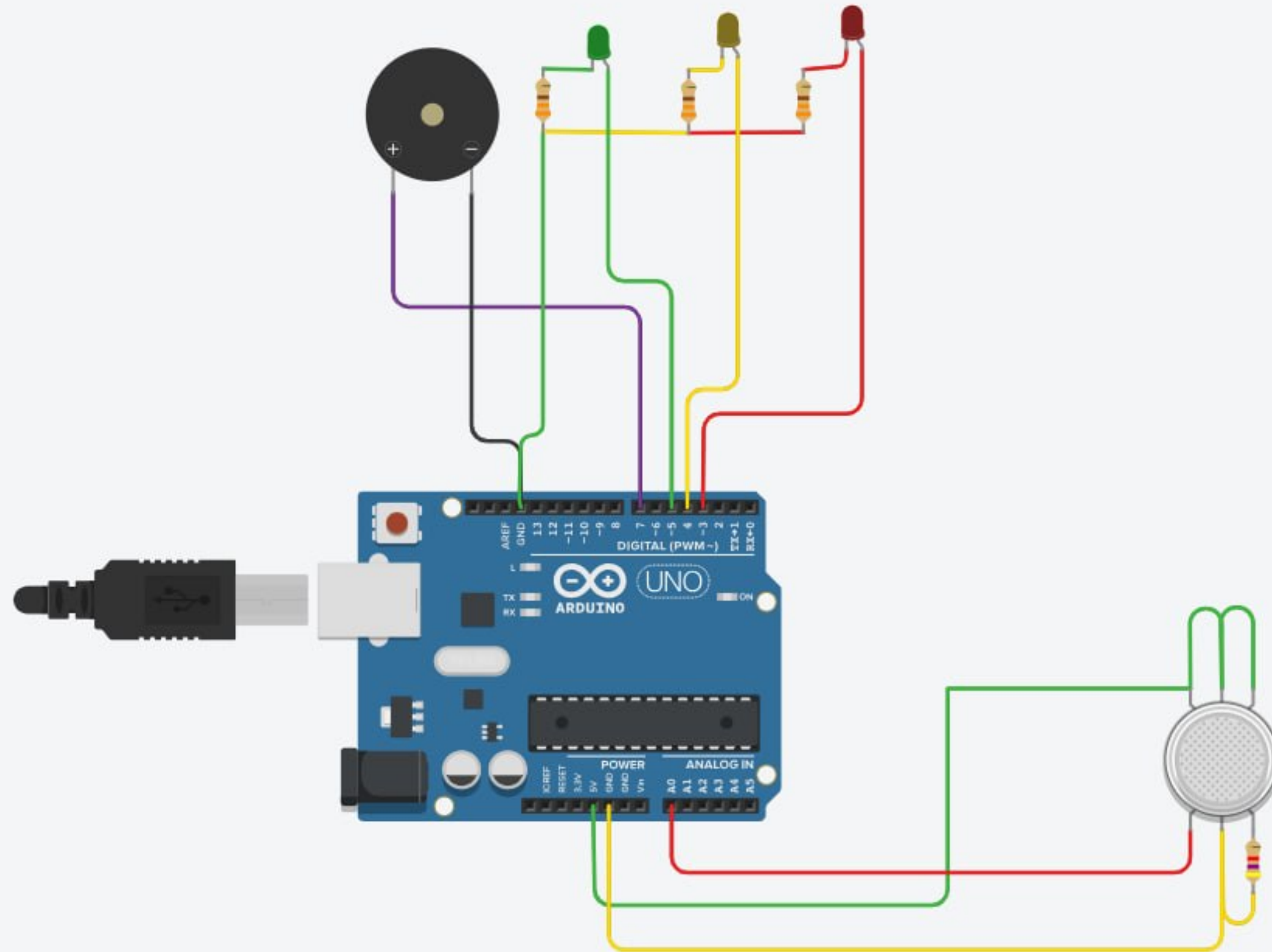
Buzzer & LED: The "Output Units" providing instant audio-visual alarms.

220 Ω Resistor: Protective component to limit current and safeguard the LED.



Breadboard & Jumpers: The foundation and connectors for a solderless, flexible prototype.

System setup & wiring



Arduino Control Logic

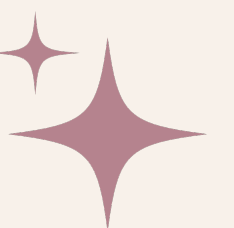
```
int gasSensor = A0; // Set CO sensor pin
int green=7, yellow=5, red=3, buzzer=8; // Set output pins
int lowT = 185, midT = 220; // Alert thresholds
```

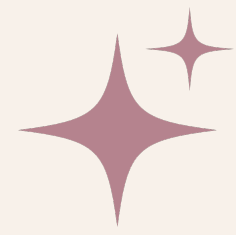
```
void setup() {
  pinMode(green, OUTPUT); pinMode(yellow, OUTPUT);
  pinMode(red, OUTPUT); pinMode(buzzer, OUTPUT);
  Serial.begin(9600); // Initialize monitor
}
```

```
void loop() {
  int gasValue = analogRead(gasSensor); // Read CO levels
```

```
    if (gasValue < lowT) {
      // Condition: Safe Atmosphere
      digitalWrite(green, HIGH);
      digitalWrite(yellow, LOW);
      digitalWrite(red, LOW);
      digitalWrite(buzzer, LOW);
    }
```

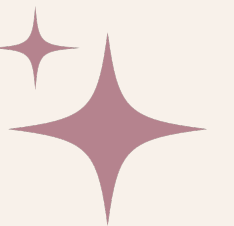
```
    else if (gasValue < midT) {
      // Condition: Early Warning
      digitalWrite(green, LOW);
      digitalWrite(yellow, HIGH);
      digitalWrite(red, LOW);
      digitalWrite(buzzer, LOW);
    }
    else {
      // ⦿ Condition: Gas Danger
      digitalWrite(green, LOW);
      digitalWrite(yellow, LOW);
      digitalWrite(red, HIGH);
      digitalWrite(buzzer, HIGH); // Alarm Sound
    }
    delay(500); // 0.5s Refresh rate
  }
```





Future Improvements Proposal: Smart CO Protection System

- **Accurate Sensor: Distinguishes between CO and smoke to prevent false alarms.**
- **Smart Alert: Sends instant notifications to your phone and automatically calls family if there's no response.**
- **Active Protection: Uses a Relay output to automatically turn on the exhaust fan, and connects to a smart gas valve to immediately shut off the gas source.**
- **Reliable: 10-year battery life with self-calibration.**

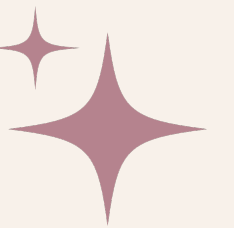




Conclusion

- **Safety First:** Our system provides 24/7 real-time monitoring for Carbon Monoxide with high accuracy.
- **Dual Alerting:** Combines visual (LEDs) and audible (Buzzer) signals to ensure immediate response.
- **Cost-Effective:** An affordable and reliable engineering solution for home and industrial safety.

In the end, our goal was to provide a smart engineering solution that can save lives from a silent killer like Carbon Monoxide. Thank you for your time, and we are now ready for your question



Thank You

